

PART 4: SEQUENTIAL LOGIC

INTRODUCTION TO SEQUENTIAL LOGIC

SAAD BAIG – DIGITAL LOGIC DESIGN

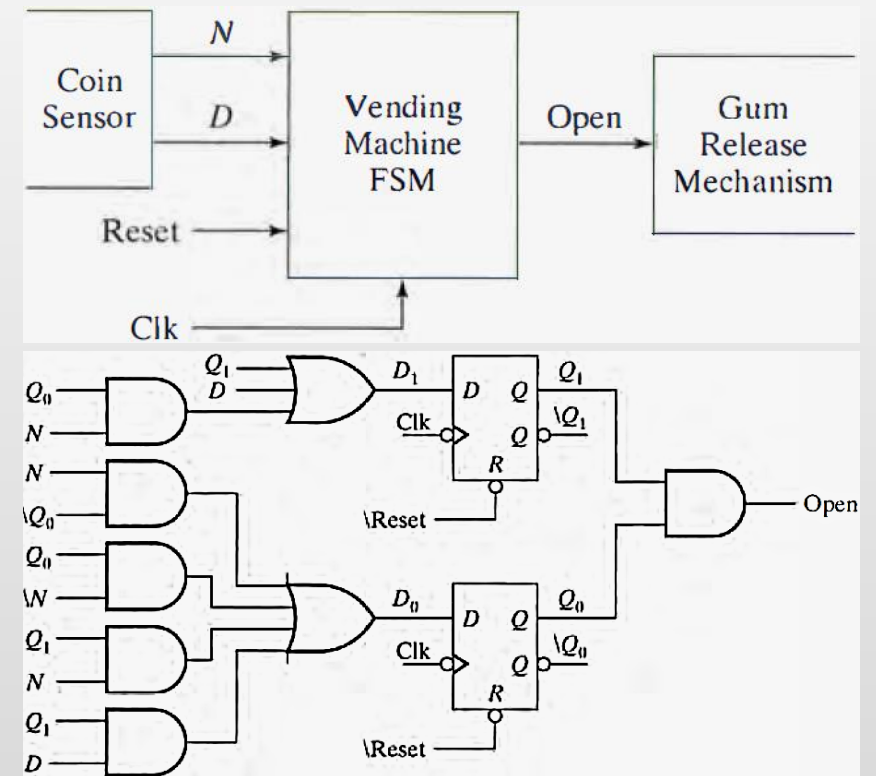


The animations and contents in these slides are originally adapted from the DLD course slides designed by Dr. Hasan Baig and Mr. Junaid Ahmed Memon. The original slides can be accessed from <https://www.hasanbaig.com/resources>

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INTRODUCTION TO SEQUENTIAL CIRCUITS

- Hand-held devices, cell phones, navigation receivers, personal computers, digital cameras, personal media players, and virtually all electronic consumer products have the ability to send, receive, store, retrieve, and process information represented in a binary format.
- The technology enabling and supporting these devices is critically dependent on electronic components that can store information, i.e., have **memory**.





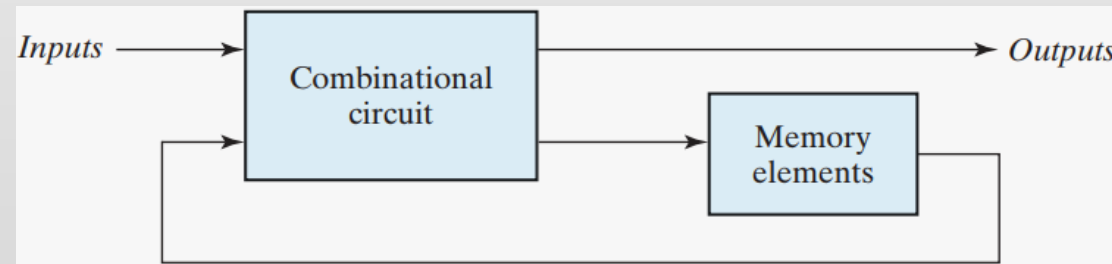
INTRODUCTION TO SEQUENTIAL CIRCUITS

- The digital circuits considered thus far have been combinational.
 - Their output depends only and immediately on their inputs.
 - They have no memory, i.e., dependence on past values of their inputs.
- Sequential circuits, however, act as storage elements and have memory. They can store, retain, and then retrieve information when needed at a later time.

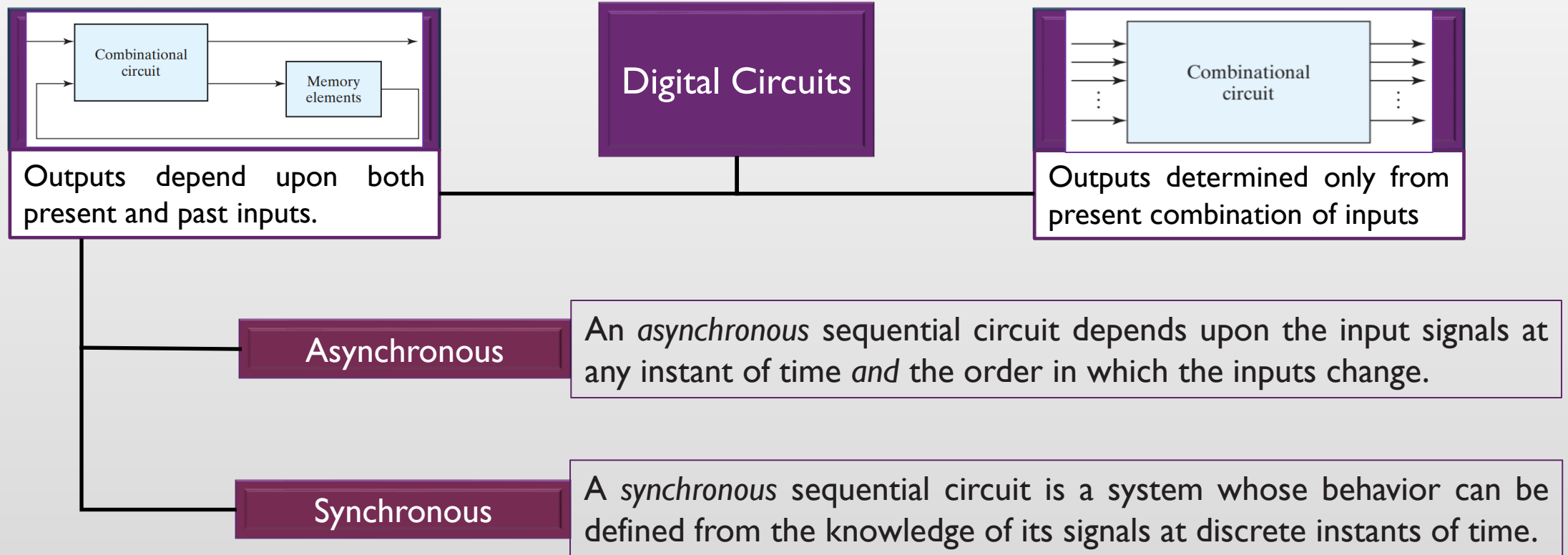
SEQUENTIAL CIRCUITS

A sequential circuit is specified by a time sequence of inputs, outputs, and internal states.

- Sequential circuit = combinational circuit + storage elements in a feedback path.
- The outputs in a sequential circuit are a function not only of the inputs, but also of the present **state** of the storage elements.
- The next state of the storage elements is also a function of external inputs and the present state.



SEQUENTIAL CIRCUITS





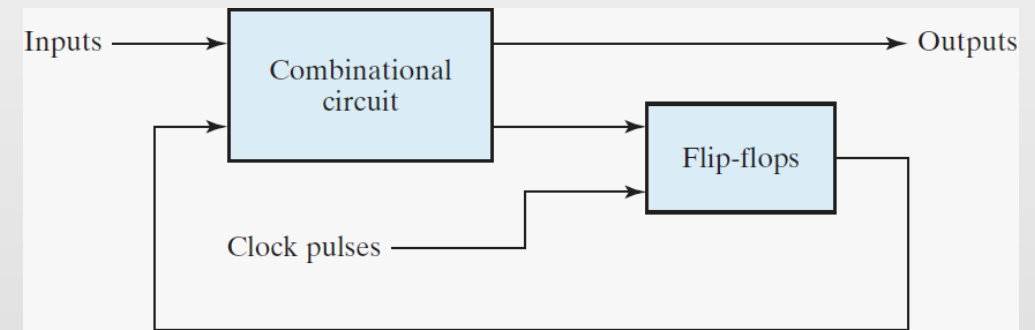
ASYNCHRONOUS SEQUENTIAL CIRCUIT

- The storage elements commonly used in asynchronous sequential circuits are time-delay devices. The storage capability of a time-delay device varies with the time it takes for the signal to propagate through the device.
- In gate-type asynchronous systems, the storage elements consist of logic gates whose propagation delay provides the required storage.
- Because of the feedback among logic gates, an asynchronous sequential circuit may become unstable at times. The instability problem imposes many difficulties on the designer.

SYNCHRONOUS SEQUENTIAL CIRCUITS

Synchronous sequential circuits that use clock pulses to control storage elements are called clocked sequential circuits.

- Synchronization is achieved by a timing device called a **clock generator**, which provides a clock signal having the form of a periodic train of clock pulses.
- The clock signal is commonly denoted by the identifiers *clock* and *clk*.



(a) Block diagram



(b) Timing diagram of clock pulses

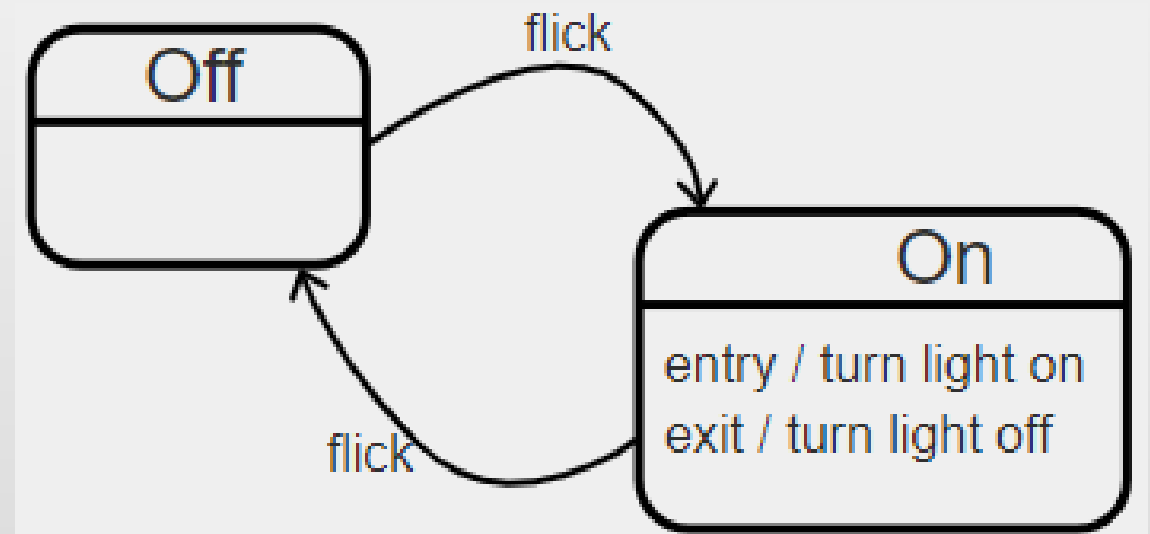


FINITE STATE MACHINE

- Sequential circuits act as storage elements and have memory. They can store, retain, and then retrieve information when needed at a later time.
- The sequential circuit is always dependent upon the **state** of a memory unit.
- When we encapsulate all information needed to determine the current value of output in a few bits then those bits are called state of system.
- Depending on the state of a memory element, it may have a 1 or a 0 at its output. We can even set the values of a memory element according to our choice.
- **Finite state machines** (or **FSMs**) are named so because the sequential logic that implements them can be in one of a fixed number of possible states.

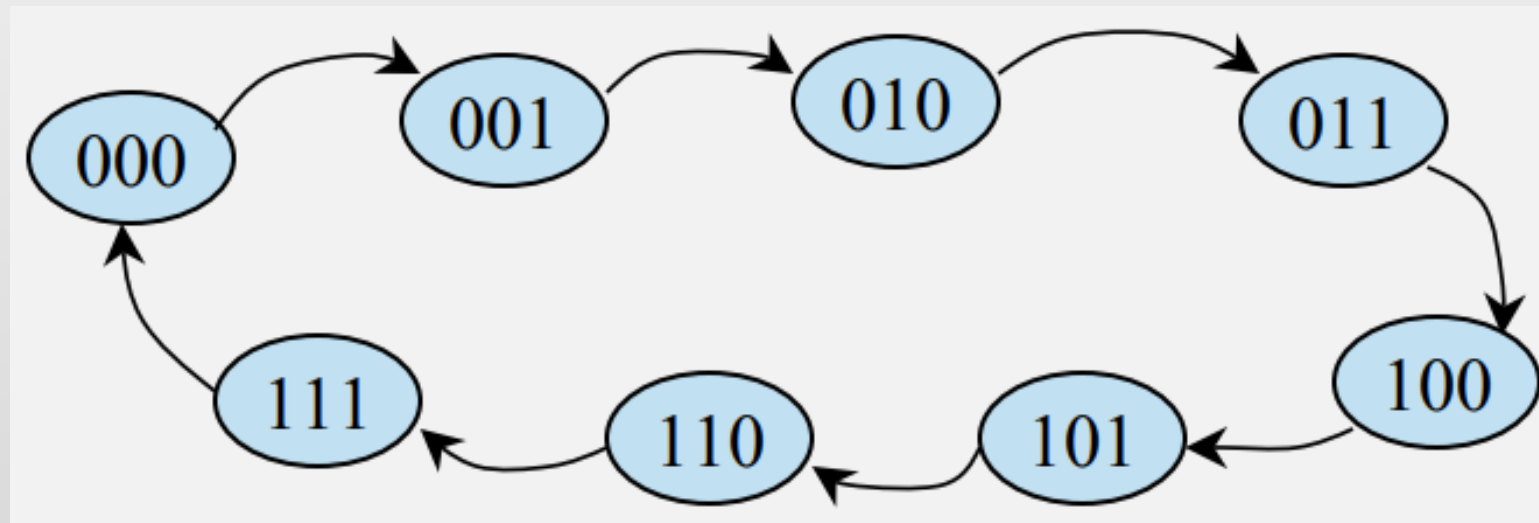
A SIMPLE STATE MACHINE

- It consists of two states, “on” and “off”.
 - This machine can therefore be in exactly one of the two states at any point in time.
 - The transitions between states are instantaneous.
- The “flick” event causes it to transition between states.
- When the machine *enters* the “on” state, a light is turned on.
- When the machine *exits* the “on” state, a light is turned off.

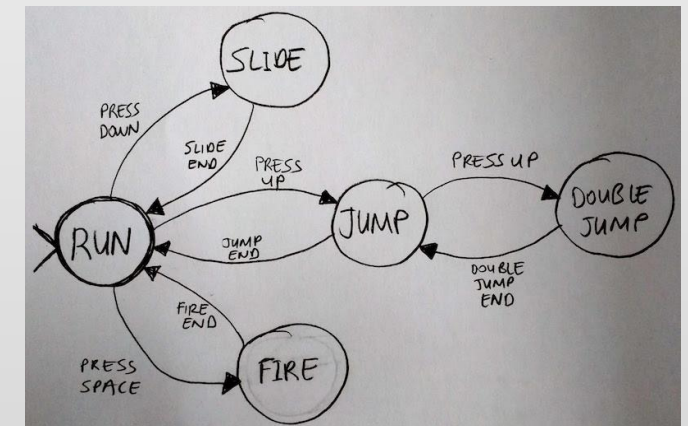
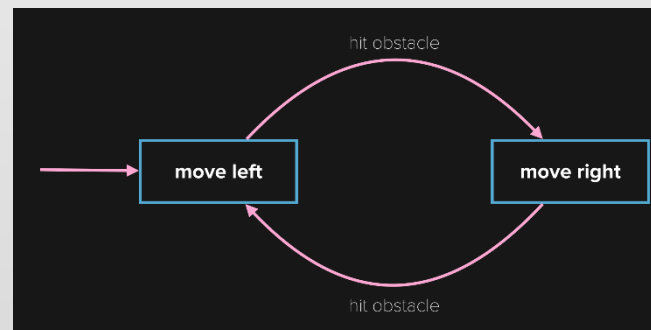
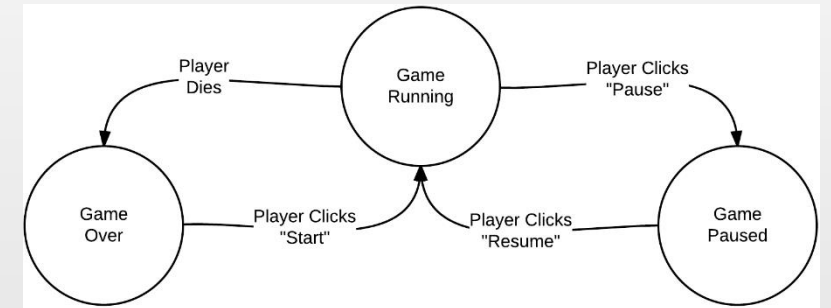
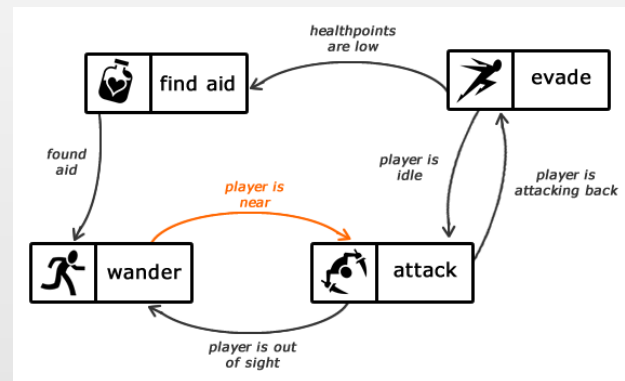
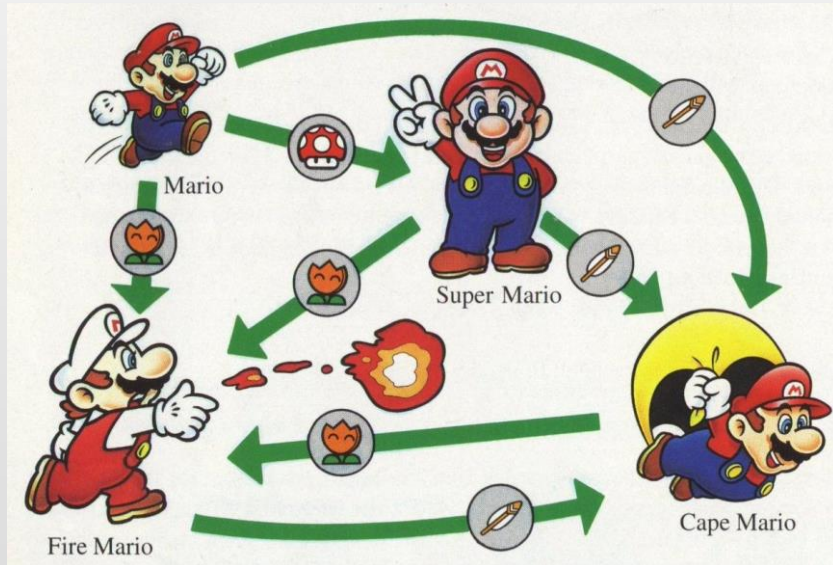


COUNTER STATE MACHINE

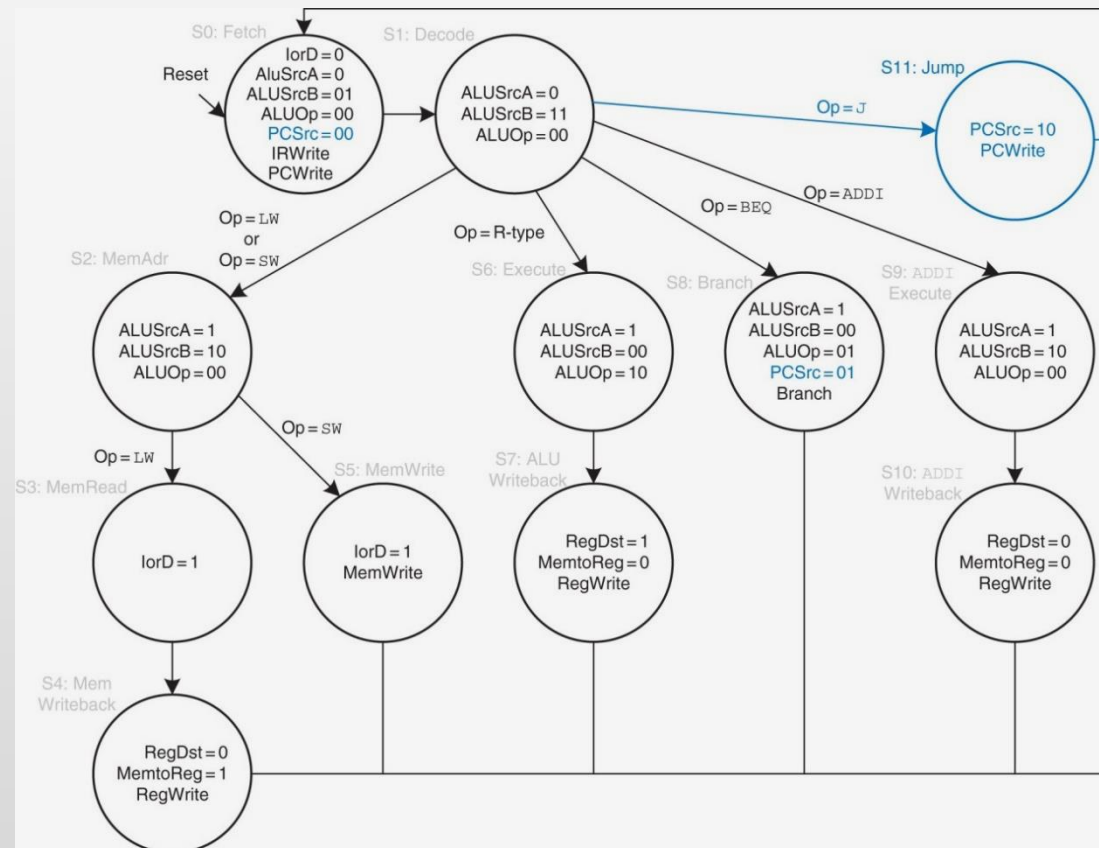
- Each clock cycle causes the memory to record a different value (i.e. proceeding from state to state).



VIDEO GAMES



INSTRUCTION CYCLE FSM



FSMS ARE EVERYWHERE!

